DD-T-90D
November 25, 1971

SUPERSEDING
Fed. Spec. DD-T-90C
July 31, 1963

FEDERAL SPECIFICATION

TABLEWARE, GLASS (BOWLS, CUPS, DISHES, PLATES, PLATTERS, AND SAUCERS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers glass tableware.

1.2 Classification. Glass tableware shall be of the items and articles listed in table I of this specification (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 406 - Plastics, Methods of Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 7350

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Items. The items shall be of the dimensions, weights, and capacities shown in table I.

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3.1.1 Material. Material shall be glass, treated to increase its resistance to mechanical shock. The ware shall have a white opal body.

3.2 Design and shape. The glass tableware shall conform to the designs and shapes shown in items Nos. 1 through 27 of this specification, as applicable, with dimensional tolerances, as indicated in 3.3.1.

3.3 Dimension, capacity, and weight. The tableware shall conform to the dimension, capacity, and weight of the specified item of tableware as shown in table I and on item Nos. 1 through 27, as applicable.

3.3.1 Dimensional tolerances. Nominal dimensions are shown in table I. Tolerance shall be allowed on those dimensions as follows:

(a) Outside top diameter:
Item Nos. 1 through 7, Nos. 10 through 23, and No. 27, plus or minus 1/16 inch.
Item No. 26, plus or minus 1/8 inch.

(b) Diameter of base:
Plus or minus 1/16 inch.

(c) Overall height:
Item Nos. 1 through 17, and Nos. 22 through 27, plus or minus 1/16 inch.

(d) Center bottom thickness:
Item Nos. 1 through 6 and Nos. 10 through 26, plus 3/64 inch or minus 1/32 inch.
Item No. 27, plus or minus 1/64 inch.

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TABLE I. Dimensions, capacity, and weight

Item No.	Article	Outside top diameter Inches	Diameter of base Inches	Overall height Inches	Minimum capacity Fluid Ounces	Center bottom thickness Inches	Maximum weight Ounces
1.	Bowl, eating	4-7/8	2-3/4	1-13/16	8.4	11/64	10-1/2
2.	Bowl, eating	5-5/8	3-5/64	2-1/8	14.0	5/32	13-3/4
4.	Dish, food, serving, (salad)	7-1/2	2-3/4	2-1/2	-	11/64	16-3/4
5.	Dish, food serving, (salad)	8-3/4	3-1/2	3-1/16	-	3/16	23
6.	Cup, drinking	3-13/16	1-19/32	2-5/16	6.5	3/16	8
7.	Cup, drinking	3-19/32	2	3-25/64	10.5	7/32	13
10.	Plate, eating, (grapefruit)	6-1/2	3-3/16	1-15/32	-	11/64	12-1/4
11.	Dish, eating, (fruit, sauce)	4-3/4	2-1/4	1-3/16	4.0	11/64	7-1/2
12.	Cup, drinking, (mug)	3-3/32	2-1/2	3-1/8	6.5	7/32	10-1/2

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TABLE I. Dimensions, capacity, and weight (con.)

Item No.	Article	Outside top diameter Inches	Diameter of base Inches	Overall height Inches	Minimum capacity Fluid Ounces	Center bottom thickness Inches	Maximum weight Ounces
14.	Plate, eating, (bread and butter)	5-1/2	3-7/32	11/16	-	11/64	8
15.	Plate, eating, (bread and butter)	6-3/4	3-29/32	51/64	-	11/64	12-1/2
16.	Plate, eating, (3 compartment)	9-1/2	7	13/16	-	11/64	24-1/2
17.	Plate, eating (dinner)	9	5-9/32	31/32	-	11/64	22
22.	Platter, food serving	12-1/2x9	9-1/4 x5-3/4	15/16	-	3/16	30
23.	Saucer, cup	5-7/8	2-25/32	15/16	-	5/32	9-1/4
26.	Plate, eating, (soup, rim)	9-1/8	4-1/2	1-5/8	-	3/16	23-1/2
27.	Dish, eating (sauce)	5-1/2	2-35/64	1-1/2	7.0	11/64	-

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3.4 Physical requirements. Tableware shall conform to the physical requirements shown in table II when tested as specified in 4.4.1, 4.4.2, 4.4.3, and 4.4.4.

TABLE II. Physical requirements

Item No.	Article	Rim warpage	Bottom warpage (thickness)	Impact strength	Drop strength	Breakage test number	Average chipping resistance
		Inches	Inches	Ft. lb.	Inches	Pieces	Ft. lb.
1.	Bowl, eating	-	-	.25	-	20	-
2.	Bowl, eating	-	-	.40	-	25	-
4.	Dish, food, (salad)	0.062	0.020	-	25	25	.16
5.	Dish, food, (salad)	.062	.020	-	50	30	.16
6.	Cup, drinking	-	-	.30	-	40	-
7.	Cup, drinking	-	-	.30	-	35	-
10.	Plate, eating (grapefruit)	.046	.020	-	25	30	.16
11.	Dish, eating, (fruit, sauce)	.031	.020	-	60	20	.16
12.	Cup, drinking, (mug)	-	-	.30	-	40	-

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TABLE II. Physical requirements (con.)

Item No.	Article	Rim warpage Inches	Bottom warpage (thickness) Inches	Impact strength Ft. lb.	Drop strength Inches	Breakage test number Pieces	Average chipping resistance Ft. lb.
14.	Plate, eating (bread and butter)	.046	.020	-	30	20	.16
15.	Plate, eating, (bread and butter)	.046	.020	-	30	25	.16
16.	Plate, eating (3 compartment)	.062	.031	-	15	40	.16
17.	Plate, eating, (dinner)	.062	.031	-	20	30	.16
22.	Platter, food serving	.062	.031	-	-	30	.16
23.	Saucer, cup	.046	.020	-	50	20	.16
26.	Plate, eating, (soup, rim)	.062	.031	-	40	30	.16
27.	Dish, eating (sauce)	.046	.020	-	25	30	.16

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3.5 Surface abrasion. Tableware shall show no signs of rupture, cracking, or chipping when tested as specified in 4.4.5.

3.6 Thermal shock resistance. Tableware shall be capable of withstanding the thermal endurance test as specified in 4.4.6.

3.7. Warpage. Rims and bases of all tableware shall be sufficiently plane so that the ware will not rock appreciably on a smooth surface. The ware shall not show evidence of warpage greater than that shown in table II (when indicated), when placed rim downward or base downward on a flat surface and held firmly in place by exerting pressure with one finger at the center of the ware so that a feeler gauge of a thickness specified in table II (when indicated) cannot be inserted at any point between the ware and the supporting flat surface.

3.8 Marking for identification. The tableware shall be permanently and distinctly marked on the bottom with the name of the manufacturer or the brand name or trademark of such known character as to be readily identified with the manufacturer.

3.9 Decoration of ware. The ware shall be decorated with blue bands. The blue color shall be in accordance with color No. 15193 of Fed. Std. No. 595. The bowls and cups, items 1, 2, 6, 7 and 12, shall be decorated with two bands on the outer surface, the first band approximately 1/8 inch wide located approximately 1/4 inch below the top edge and the second band approximately 1/32 inch wide located approximately 5/8 inch below the first band on items 1, 2, 6 and 12 and approximately 3/4 inch below the first band on item 7. The plates and platter, items 14, 15, 16, 17, 22 and 26, shall be decorated with two bands on the upper surface, the outer band approximately 1/8 inch wide located approximately 1/4 inch from the outer edge and the inner band approximately 1/32 inch wide located approximately 3/16 inch above the indentation. The salad dishes and grapefruit plate, items 4, 5 and 10, shall be decorated with two bands on the upper surface of the rim, the outer band approximately 1/8 inch wide located approximately 3/16 inch from the outer edge, and the inner band approximately 1/32 inch wide located approximately 1/4 inch from the inner edge of the rim. The sauce dishes and saucer, items 11, 27 and 23 shall be decorated with one band approximately 1/8 inch wide located on the upper surface approximately 3/16 inch from the outer edge.

3.10 Workmanship. The tableware shall be uniformly constructed, free from cracks, chips, blisters, stones, or other imperfections which may impair their usefulness. The edges shall be well rounded and sufficiently fire-polished to remove fins and sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Sampling for inspection and tests. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling a lot for inspection and tests shall consist of all material of the same item, design, and shape of tableware made by the same process from the same components by one manufacturer and submitted for inspection at the same time.

4.2 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used in the end item were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. Examination of the end item shall be in accordance with the classification of defects in applicable sub-paragraphs below and at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.4. Random samples shall be drawn from each lot of end items of the same design, shape, dimension, weight, capacity, and color for examination of visual, dimensional, count, and preparation for delivery defects. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of tableware of the specified item, for the examination in 4.3.1.1 and 4.3.1.2, and in units of shipping containers for examination under 4.3.1.3.

4.3.1.1 Examination of the end item for defects in appearance, construction, design, and general workmanship. The end item shall be examined for defects in table III. The sample unit for this examination shall be one complete item of tableware.

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TABLE III. Visual defects

Examine	Defect
Appearance	Not item, shape, or design specified. Not a white opal body. Not clean, any discoloration. Mottled areas; visible particles of foreign or unmelted material.
Workmanship	Nick, chip, fracture, or crack (including firecracks); heavy lines, waves, strings, cords, or shear marks; scratches or indentations, which are visible at a distance of 3 feet. Open or soft blisters. Stones (undissolved material), greater than 3/32 inch. Blisters (solid or closed)-maximum size 3/32 inch. Blisters (soft or open)-all degrees. Rack marks (small notch on foot)-marks over 1/8 inch in width or length. (Discoloration of rack marks is not a defect). Grind or scuff marks, rough or abraded surface; sags, runs, flow, or wrinkles, which are visible at a distance of 3 feet.
Construction	Varies from design and shape shown in the applicable drawing for the specified item. Corners and edges not blended smoothly; outer edges not well rounded; sharp corners or edges. Roll or bead (as applicable) under outer edge of flatware, omitted. Handles or handle openings omitted, distorted, or not in position shown in the applicable item drawing; not smoothly blended into body. Surfaces and curves not blended smoothly. Foot rings (as required) omitted, or not blended smoothly into body. Item of tableware does not rest evenly and firmly when placed in normal use position on a flat surface or table, or the rim or bottom warpage exceeds the permissible tolerances indicated in table II (as applicable). Identification markings omitted, illegible, incorrect, or not as specified; not on outside bottom surface, unless specified (see 3.8).

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TABLE III. Visual defects (con.)

Examine	Defect
Decorations (blue bands)	Not color specified; not uniform throughout, any feathering ragged edges, runs, mottle, or spots, blur or smear; not in location specified.

4.3.1.2 Examination of the end item for defects in dimensions, capacity, and weight, as applicable. The end item shall be examined for defects in table IV. The sample unit for this examination shall be one complete item of tableware.

TABLE IV. Dimensions, capacity, and weight

Examine	Defect
Dimensions	Any dimension exceeding size specified for the applicable item by more than the permissible tolerances as indicated in table I and 3.3.1, or the applicable item figure.
Capacity (as applicable)	Less than the minimum capacity indicated in table I for the applicable item of tableware
Weight	Greater than the maximum weight indicated in table I for the applicable item of tableware.

4.3.1.3 Examination of preparation for delivery requirements. An examination shall be made to determine that the prepacking, packing, and marking complies with the section 5 requirements. Defects shall be as indicated in table V. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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TABLE V. Examination of preparation for delivery

Examine	Defect
Prepacking	Not level specified; not in accordance with contract requirements. Unit items not prepacked as specified. Unit items not snugly packed; box liner, partitions, pads, or cushioning material (as applicable) omitted or not type material specified. Arrangement or number of unit items per container not in accordance with contract requirements. Mixed items, designs, and shapes in same container. Container material not as specified; closures not accomplished by specified or required methods or materials.
Packing (as applicable)	Not level specified; not in accordance with contract requirements. Mixed prepacked items in same container. Box liner material (as applicable) omitted, damaged, or not as specified. Container material not as specified; closures not accomplished by specified or required methods or materials.
Markings (as applicable)	Exterior markings (as applicable) illegible, incorrect, incomplete, omitted, or not in accordance with contract requirements. Special markings omitted, or not as specified.

4.3.1.4 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels for determining the sample size, and the acceptable quality levels (AQLs) expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQLs
4.3.1.1	I	2.5
4.3.1.2	S-2	4.0
4.3.1.3	S-1	2.5

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4.3.2 Testing of the end item. Each end item lot shall be tested for the applicable characteristics specified in table VI. The sample unit shall be 7 pieces of tableware selected from one shipping container of the same item. No more than one test outlined in table VI shall be performed on any one piece of tableware and no more than seven pieces of tableware shall be selected from any one shipping container. The inspection level shall be S-1 and the acceptable quality level (AQL) shall be 6.5 expressed in defects per hundred units. The failure of any one piece of tableware shall be cause for rejection of the entire sample unit. All test reports shall contain the individual values utilized in expressing the final values.

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TABLE VI. End item testing

Characteristic	Specification reference		Requirements applicable to		Number determinations per sample unit	Results reported as	
	Requirement	Test method	Sample unit	Lot aver.		Pass or fail 1/	Numerically to nearest
Impact strength	Table II	4.4.1					
Rim			X	-	4	X	-
Sidewall			X	-	4	X	-
Drop strength	Table II	4.4.2	X	-	3	X	-
Breakage	Table II	4.4.3	X	-	1	X	-
Chipping resistance	Table II	4.4.4	-	X	Average of 3	-	0.01 foot pounds
Surface abrasion	3.5	4.4.5	X	-	1	X	-
Thermal shock	3.6	4.4.6	X	-	1	X	-

1/ If failure is indicated, report either description or numerical point of failure, as applicable.

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4.4 Test procedures.

4.4.1 Impact strength.

4.4.1.1 Apparatus. Impact strength test shall be made on the items listed in table II for which impact strength is shown, and the ware must meet the impact strength (foot poundage) requirements in that table. The ware shall be tested by using the apparatus shown in figures 1 and 2. The hammer for this test shall be a plastic ball 2-1/8 inches in diameter and weighing approximately 5 ounces. The plastic ball shall have a Rockwell hardness between 105 and 120 on the "M" scale (see method 1081 of Fed. Std. No. 406). The ball shall be attached to the steel tape by the illustrated holding cradle and the weight of the assembly shall be approximately 12 ounces. The dial of the apparatus shall be properly calibrated for the assembly used. Both faces of the "V" block shall be lined with 1/4-inch tempered pressed wood smooth side out (securely fastened to insure complete surface contract with the "V" block) having a Rockwell hardness between 65 and 95 on the "R" scale.

4.4.1.2 Procedure. Before testing, the inside surface of each specimen to be tested shall be abraded parallel to its principal (vertical) axis by means of fresh No. 150-grit silicon carbide paper or cloth, mounted on a steel cylinder 1 inch in diameter, 4 inches long. A recommended method used for abrading the ware (a) and method of testing (b) follows:

(a) The specimen is held on its side and the cylinder held lightly at one end in an approximately horizontal position and permitted to rest the major portion of its weight on the rim of the specimen. The cylinder is then moved in and out abrading the rim so as to produce scratches not more than 1/16 inch apart. During this abrading motion the specimen and cylinder are slowly rotated as the abrading continues around the rim. Normally, one or two revolutions of the specimen will suffice.

(b) The specimen, resting on its base, shall be placed between the two cast iron blocks, lined with tempered pressed wood as above, forming a 90-degree "V" and the apparatus shall be so adjusted that when the hammer hangs vertically, its impact point touches the article at the junction of the rim curvature and side wall and in the plane of the bisector of the angle of the "V". The specimen shall be struck four blows, each blow approximately 90 degrees apart and at an energy value specified in table II. Failure in each case is defined as the development of a rupture or crack or the breaking of a portion of the ware.

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4.4.2 Drop test. The drop test shall be made on items listed in table II, for which drop strength is shown and the ware must meet the drop-strength requirement in that table. Prior to dropping, each specimen to be tested shall be abraded by hand on the entire inner and outer surfaces using fresh No. 150-grit silicon-carbide paper or cloth. Special care shall be given to abrading the inner and outer well portions and rim of the specimen. Care shall be taken that the specimen is abraded in all directions in such a manner that the scratches, even though small and difficult to see, will be a maximum of 1/16 inch apart. Specimen shall then be tested by dropping onto a level, smooth maple plank 1-3/4 inches thick, approximately 14 inches wide and 20 inches long, which is supported on two wood strips, 1-3/4 inches wide by 3/4 inch thick, parallel to and 2 inches from the ends of the plank. The plank shall be supported solidly by two bolts through each of these strips to a solid concrete base or floor. The specimens shall be dropped bottom downward in such a manner so as to strike as nearly flat as possible. A vacuum chuck or other suitable dropping mechanism may be used so that ware will strike flat. Three drops shall be made on each specimen at the height specified in table II.

4.4.3 Breakage test. Breakage tests shall be run on each item of tableware. The test shall be made by placing the specimen upright on its base upon a solid support (1-3/4-inch maple plank as used for the drop test and specified in 4.4.2) placing a steel center punch (point ground to approximately a 60-degree included angle) in contact with the center of the inside surface of the bottom and striking with a hammer in successive blows of increasing severity until breakage occurs. The specimen should break into a small number of irregular-shaped pieces not greater in number than specified in table II and it must not dice. Any piece 1/4 inch or more on any three of its adjacent edges (excluding the thickness dimension) shall be included in the number counted. Smaller fragments shall not be counted.

4.4.4 Chipping resistance.

4.4.4.1 Apparatus. The chipping-resistance test shall be made on the items listed in table II, for which an average chipping resistance is shown and the ware must meet the chipping-resistance requirements in that table. The ware shall be tested by using the apparatus and 6-ounce hammer as shown in figure 2. The hammer shall be steel hardened to a Rockwell hardness between 45 and 60 on the "C" scale. The curved impact face of the hammer shall be smooth.

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4.4.4.2 Procedure. The specimen resting on its base shall be placed between the two cast-iron blocks, not lined, and the apparatus so adjusted that when the hammer hangs vertically, the center of the impact face of the chipping hammer touches the edge of the test specimen in the plane of the bisector of the angle of the "V". The specimen shall be repeatedly struck on the edge at the same point by the hammer with an initial blow of increasing force, beginning with an initial blow of 0.11 foot-pound and increasing the force of each succeeding blow in increments of 0.02 foot-pound until failure occurs. Failure consists in chipping the edge of the test specimen so that fragments are removed and the chipped portion of the rim is approximately 3/8 inch in any one dimension. The specimen shall be tested at three equally spaced points on its periphery, except that for the platter the points shall be at both ends and in the center of one side. The average energy for the three points tested shall be the value reported.

4.4.5 Surface abrasion. The surface abrasion test shall be made on each item of tableware. Each specimen shall be thoroughly abraded on the entire inner and outer surfaces using fresh No. 50-grit silicon-carbide paper or cloth. After abrading, the specimen shall be immersed in tap water for 5 minutes. The development of a rupture, or any cracking or chipping, shall constitute failure.

4.4.6 Thermal shock. Thermal-shock tests shall be made on all types of tableware. Test specimens of ware shall be heated in an oven at a temperature of 300 degrees plus or minus 5°F. for a period of not less than 30 minutes. The ware shall then be immediately plunged into a cold water bath maintained at a temperature between 32° and 40°F. Any cracking or chipping shall constitute failure.

5. PREPARATION FOR DELIVERY

5.1 Prepacking. Prepacking shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Items 1 and 2: Bowls, eating. Each shipping container shall have inserted two sets of slotted partitions, separated by a dividing pad, and so designed as to provide individual cells for each bowl. Each partition shall be designed to contain eight bowls in length and three in width. Each bowl shall be inserted on edge into the individual cells formed by the partitions. The two outside rows of eight bowls shall be placed into cells so that four bowls have their bases facing each end of the box. The middle

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row shall be placed so that the bases of all eight bowls of each end shall face toward the center of the container. In addition, each shipping container shall be fitted with a taped box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636.

5.1.1.2 Items, 4, 5 and 26: Dish, food serving, (salad); and plate, eating, (soup, rim). Each shipping container shall be designed so as to provide space for four shells (tubes) each forming a rectangle, positioned two by two and containing six identical items each. Each dish, food serving, (salad) and plate, eating, (soup rim) shall be separated by a square separator pad that is at least 1 inch greater than the maximum dimension of the item and having a thickness of not less than 0.009 inch. The separator shall be made of 21-pound minimum basis weight chipboard. The dishes and plates shall be stacked on edge with bases facing toward the center of the container.

5.1.1.3 Items 6, 7, and 12: Cup, drinking, and cup, drinking, (mug). Each shipping container shall have inserted three sets of slotted partitions, separated by two dividing pads, and designed so as to provide cells for each cup or mug. Each partition shall be designed to contain four cups or mugs in length and four in width. All cups and mugs in the bottom partition shall be placed bases down; in the middle partition bases up or down; and in the top partition bases up. In addition, each shipping container shall be fitted with a taped box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636.

5.1.1.4 Items 10, 11, 14, 15, 23, and 27: Plate, eating, (grapefruit); dish eating, (fruit, sauce); plate, eating, (bread and butter); saucer, cup; and dish, eating, (sauce). Each shipping container shall have inserted one set of slotted partitions designed so as to provide four cells, each cell to contain a stack of 12 dishes, plates, or saucers. Each dish, plate, and saucer shall be separated by a square separator pad that is at least 1 inch greater than the maximum dimension of the item and having a thickness of not less than 0.009 inch. The separator shall be made of 21-pound minimum basis weight chipboard. The dishes, plates, and saucers shall be stacked flat in the container, two diagonal stacks with bases down and two diagonal stacks with bases up. In addition, each container shall be fitted with a taped box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636.

5.1.1.5 Items 16, 17, and 22: Plate, eating, (3-compartment); plate eating (dinner); and platter, food serving. Each shipping container shall be provided with a snug-fitting corrugated fiberboard shell (tube) to contain a stack of 24 plates or platters. Each container shall be provided with a top and bottom pad and a dividing pad each cut to fit within the fiberboard shell (tube).

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Each plate and platter shall be separated by a square separator pad that is at least 1 inch greater than the maximum dimension of the item and having a thickness of not less than 0.009 inch. Each plate and platter shall be placed flat, 12 with bases up, 12 with bases down, and the bases facing the dividing pad.

5.1.1.6 Partitions, pads, and shells (tubes). All the partitions, dividing pads, top and bottom pads, and shells (tubes) shall be made from type CF (A or C flute), class domestic, variety SW, grade 275 of PPP-F-320.

5.1.2 Level C. Tableware shall be prepacked to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Tableware of one description only, prepacked with protective cellulosic material and in the quantity specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style CSSC or CSSC-L (as applicable), grade V2s of PPP-B-636. Each shipping container shall be closed, waterproofed, and reinforced in accordance with the appendix of the container specification.

5.2.2 Level B. Tableware of one description only, prepacked with protective cellulosic material and in the quantity specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style CSSC or CSSC-L (as applicable), type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.1 When specified (see 6.2), the fiberboard shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification.

5.2.3 Level C. Tableware, prepacked as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123. Each shipping container shall be printed or stenciled on the top of the container with letters at least three-quarters of an inch high as follows:

"GLASS - DO NOT DROP OR THROW"

5.3.1.1 Special marking. At least three surfaces of the shipping container shall bear a white label on which is imprinted in red a 1/4 inch border, fracture features and the word "FRAGILE". On containers for all items, except eating dish and saucer, the fragile label shall be 4-inches square and shall be so placed as to be conspicuous, but not to interfere with other markings. The label shall not be placed on the lower one-third of the sides of the container, unless specifically required by law. The label for the eating dish and saucer containers shall be 2-1/4-inches square and the preceding restriction on the placement of the label shall not apply.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2.1 Special marking. Each shipping container shall be provided with fragile labels in accordance with MIL-STD-129. In addition, the top panel of each container shall be printed or stenciled in black ink with capital letters not less than three-quarters of an inch in height as follows:

"GLASSWARE
HANDLE WITH CARE"

6. NOTES

6.1 Intended use. The glass tableware is intended for use by civilian and military agencies in preparing, serving, and storing food.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Items required (see 1.2 and table I).
(c) Selection of the applicable levels of prepacking and packing required (see 5.1 and 5.2).
(d) When weather resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

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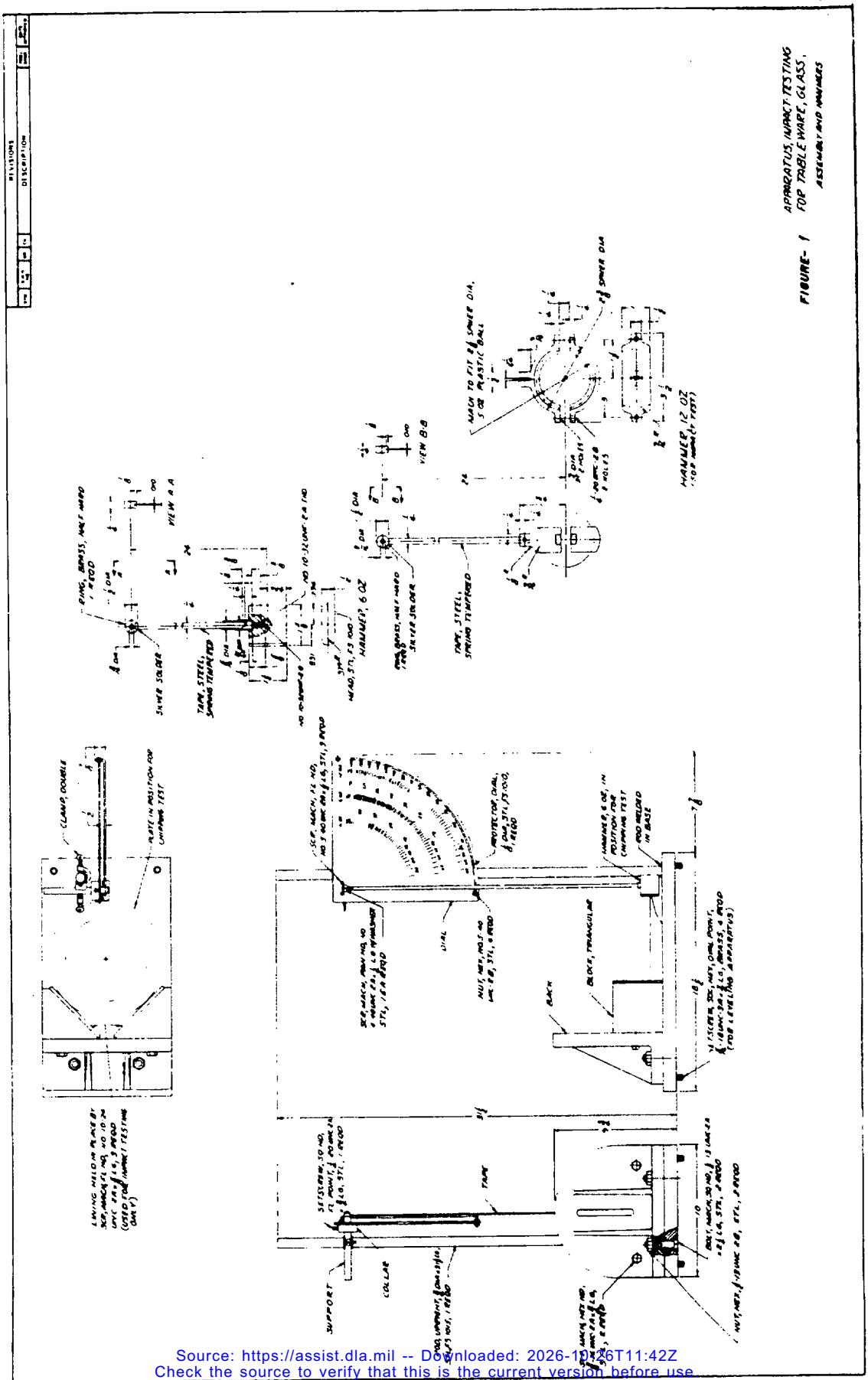


FIGURE 1 APPARATUS, IMPACT TESTING FOR TABLEWARE, GLASS; ASSEMBLY AND IMAGES

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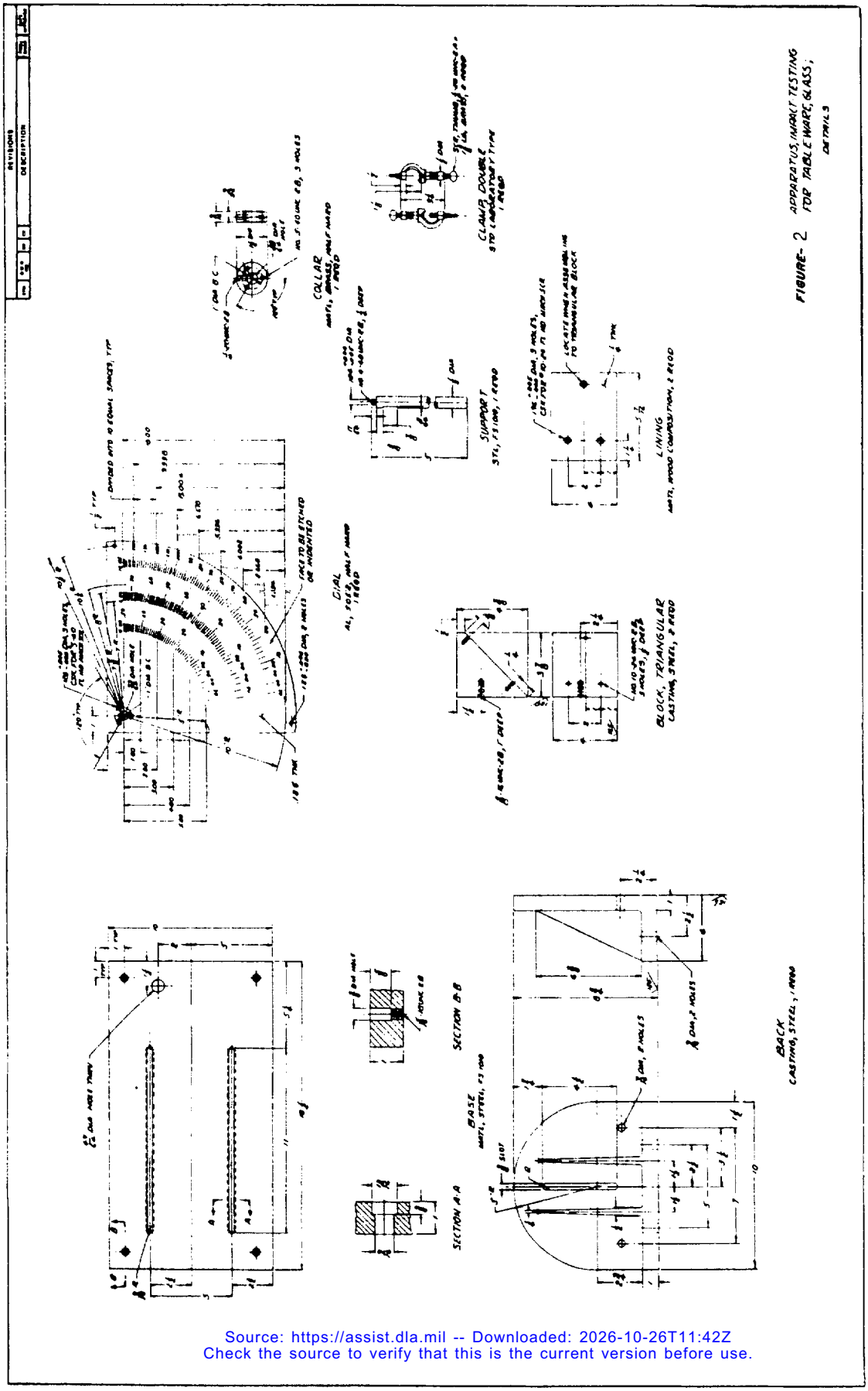
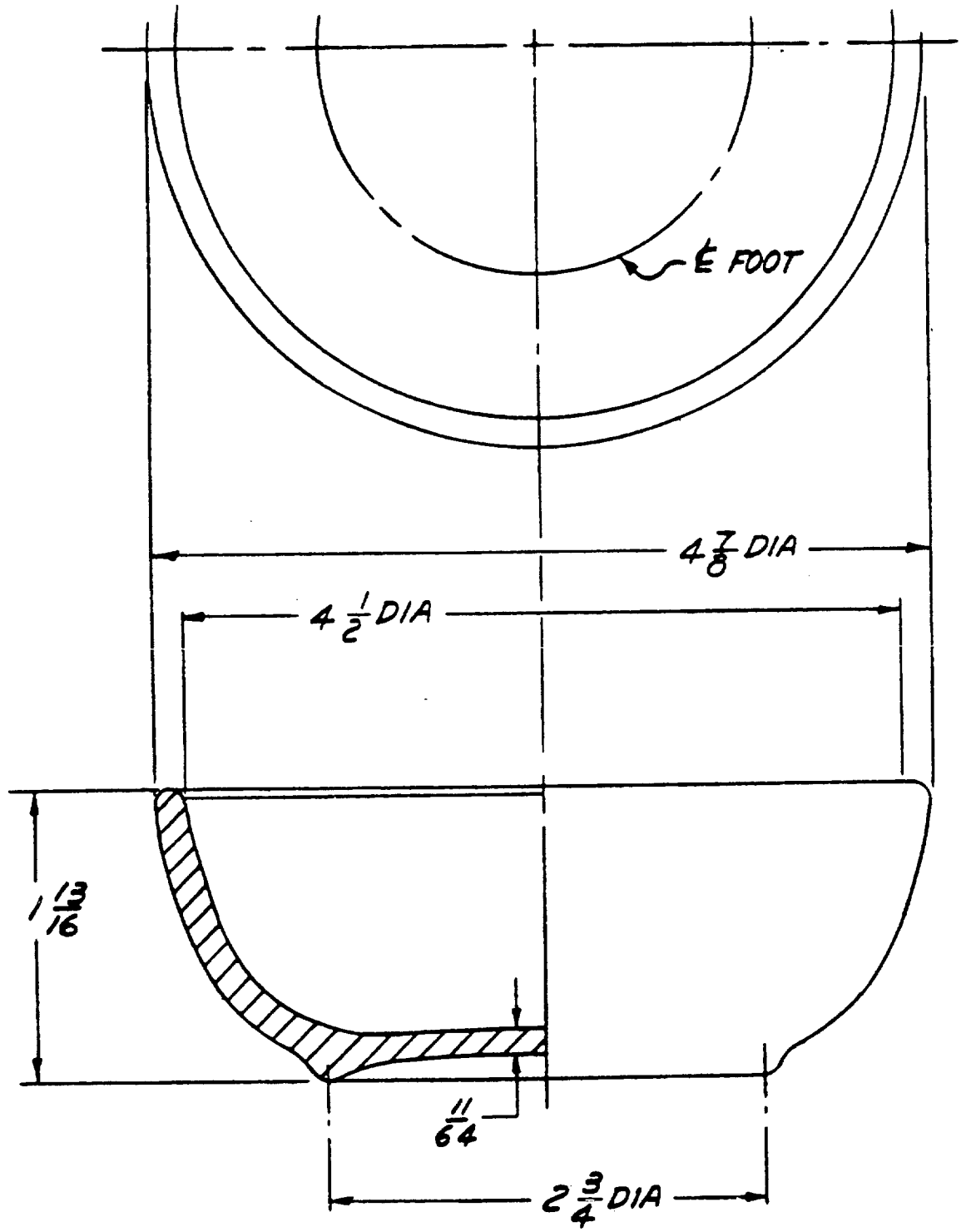


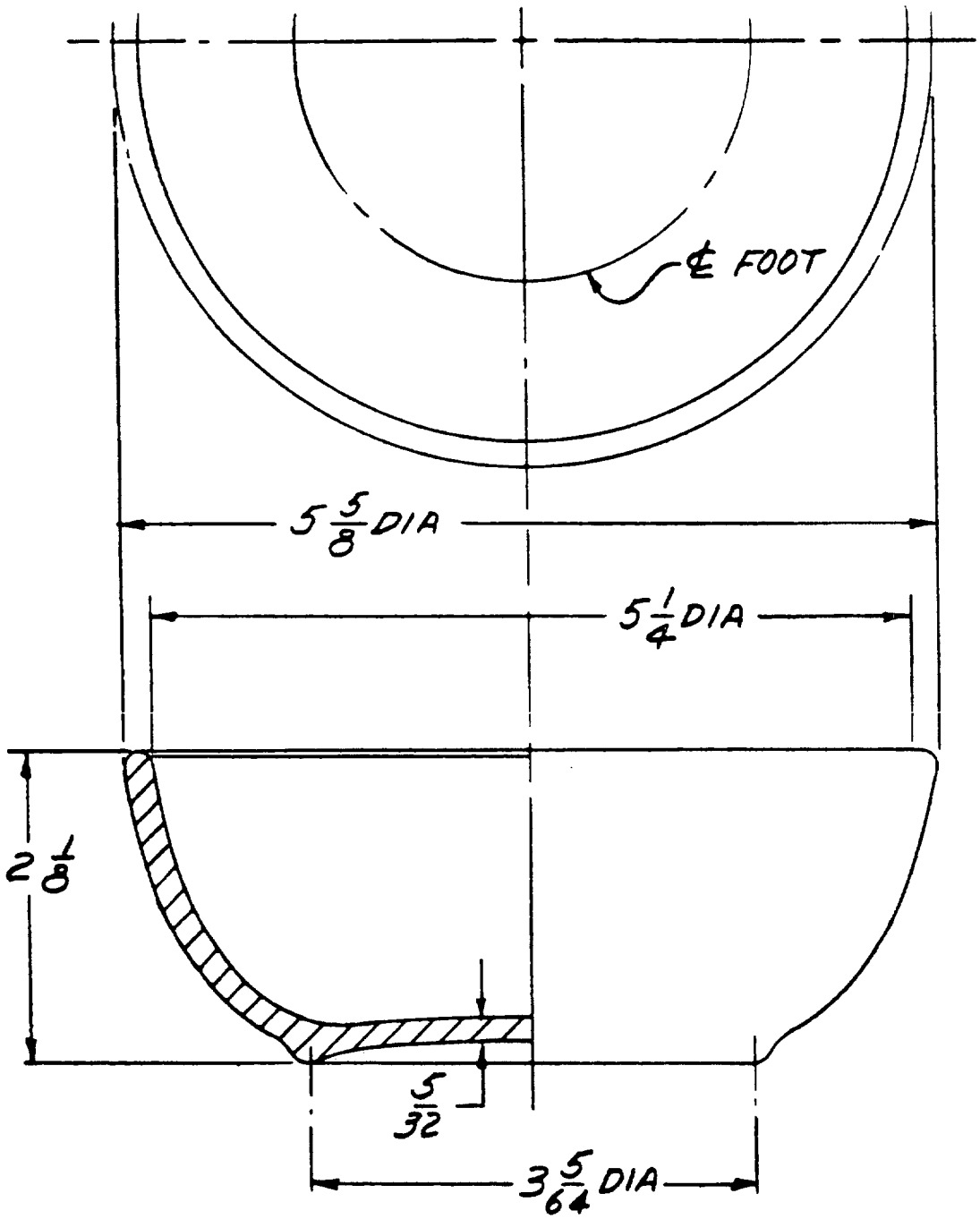
FIGURE 2 APPARATUS, IMPACT TESTING FOR TABLE WARE, GLASS; DETAILS

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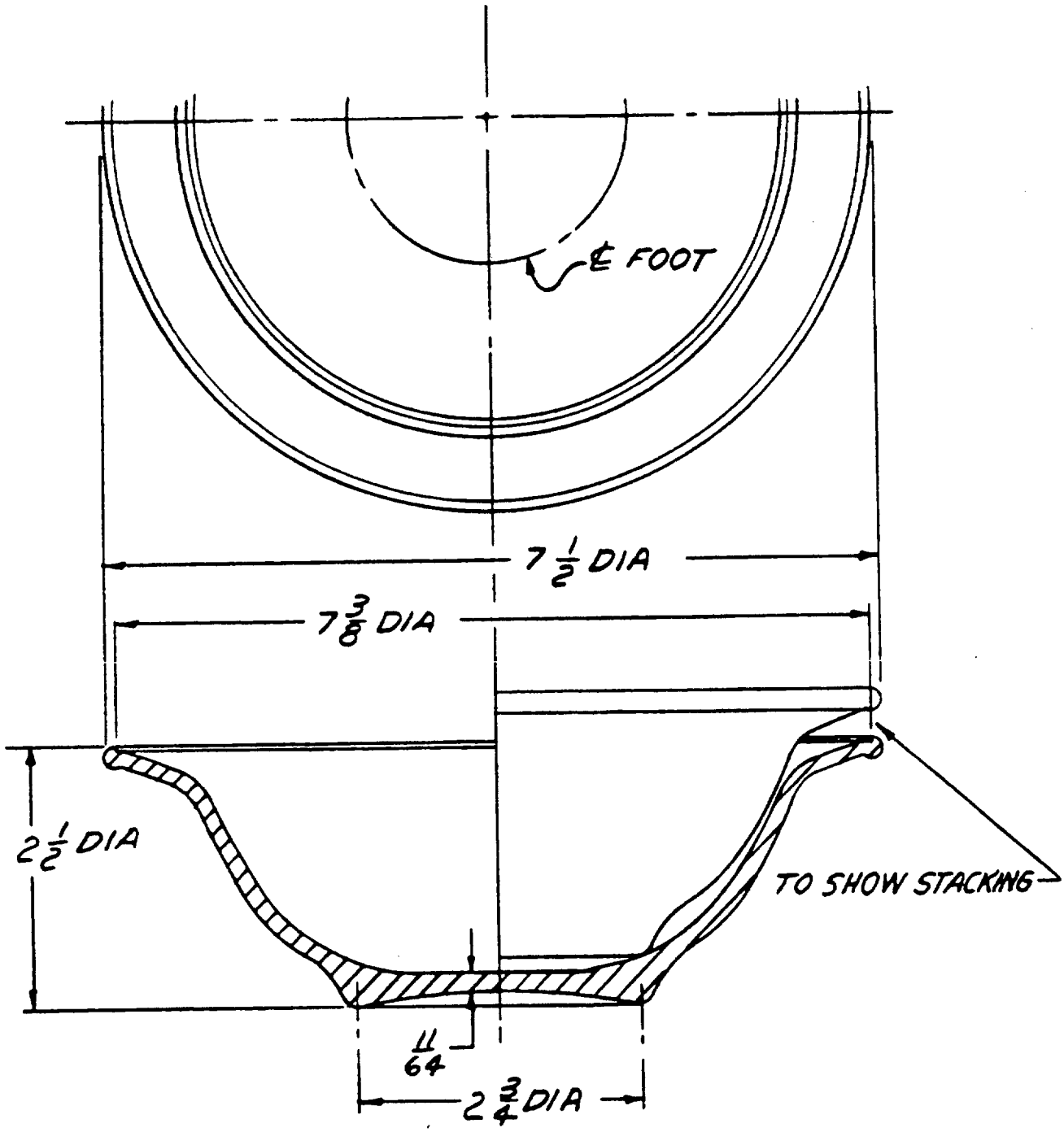
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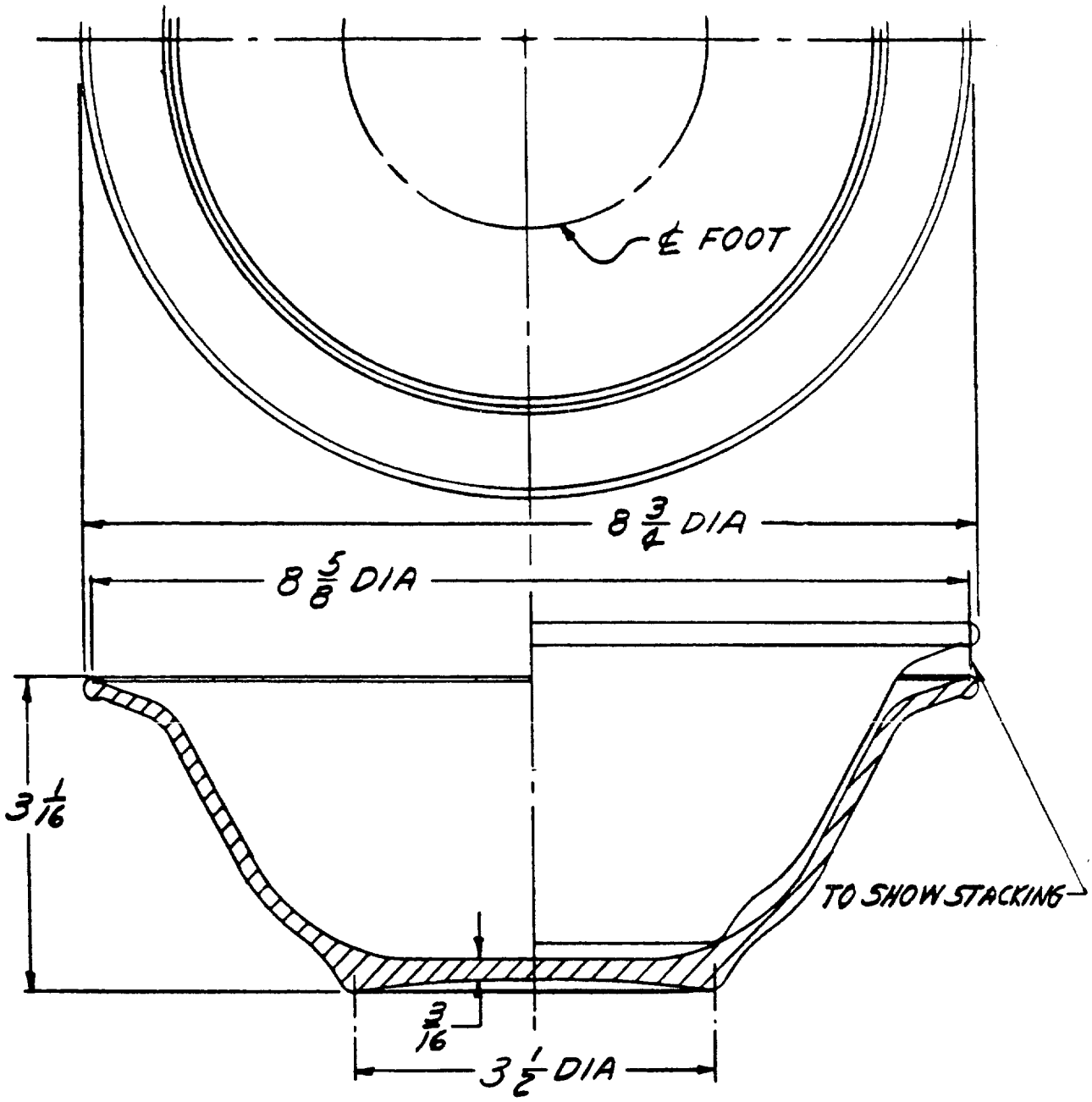
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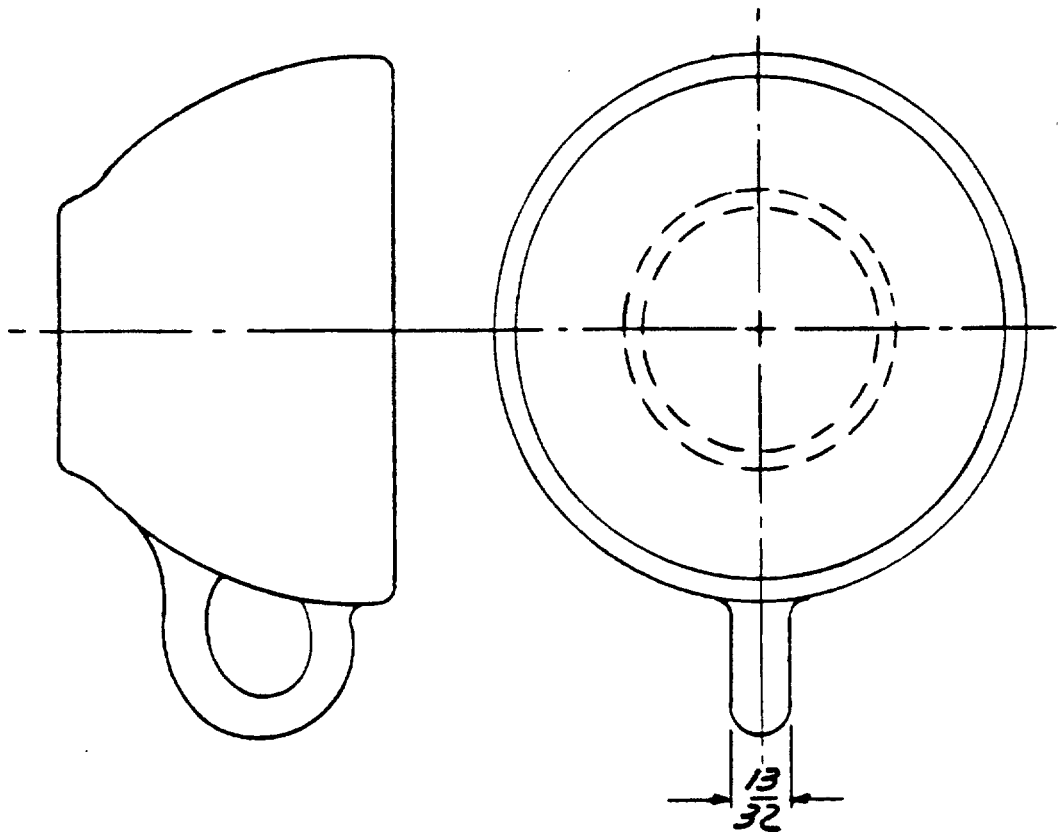
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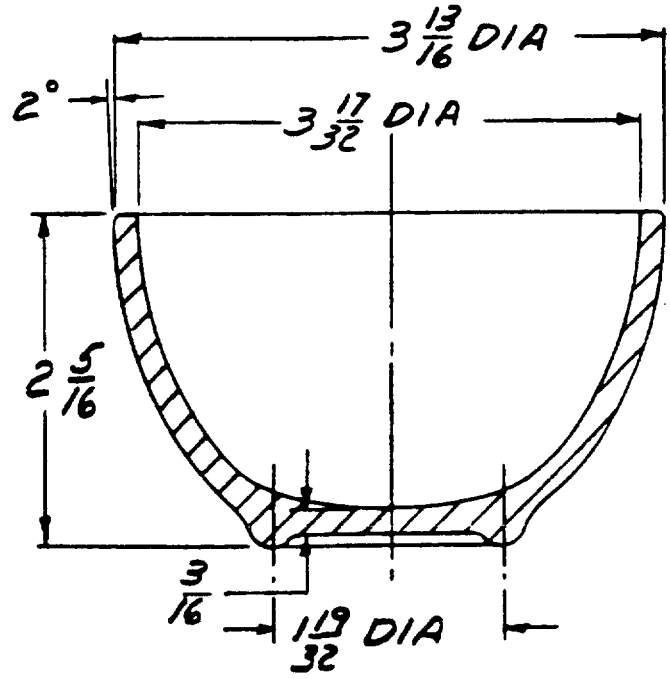
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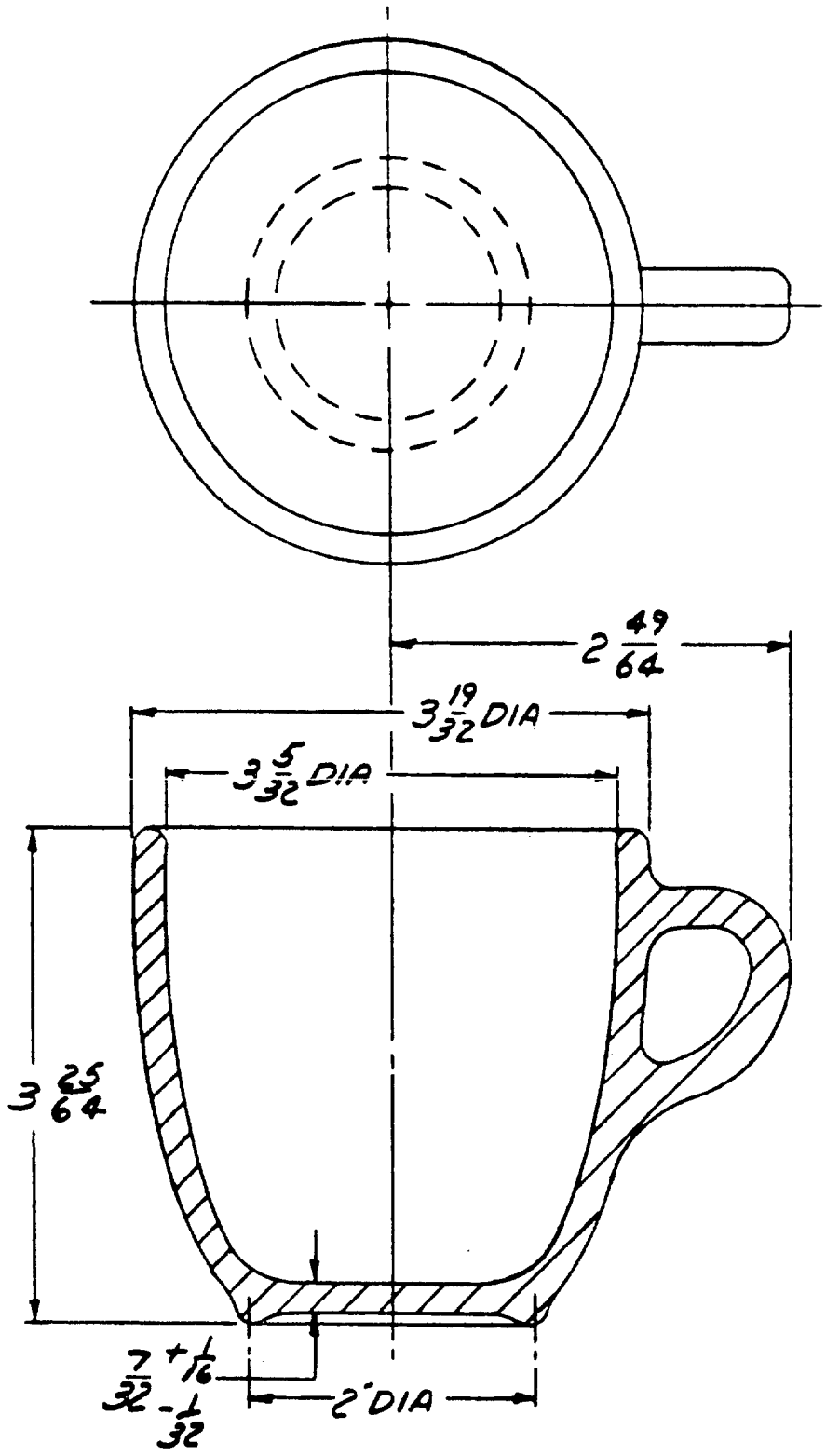
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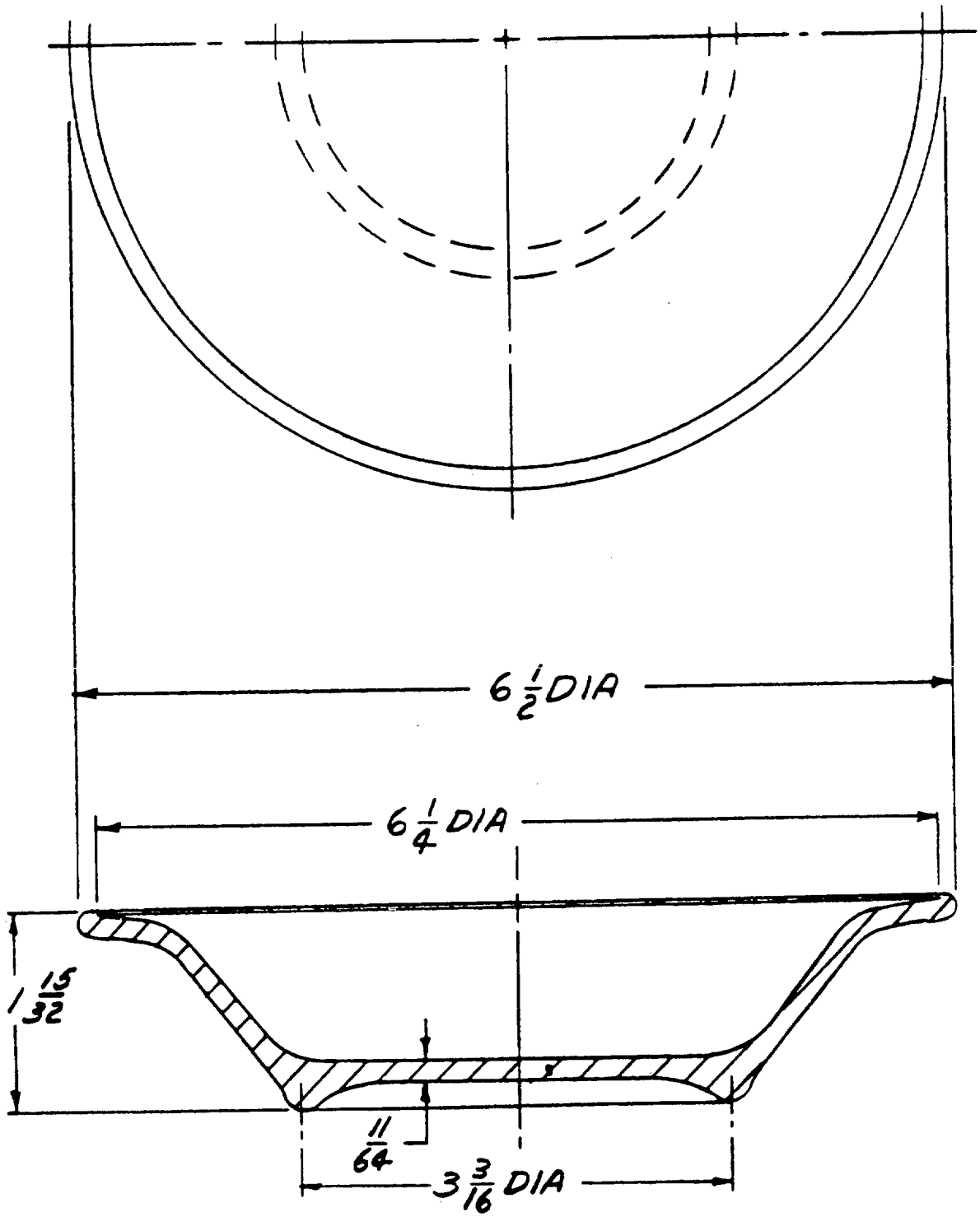


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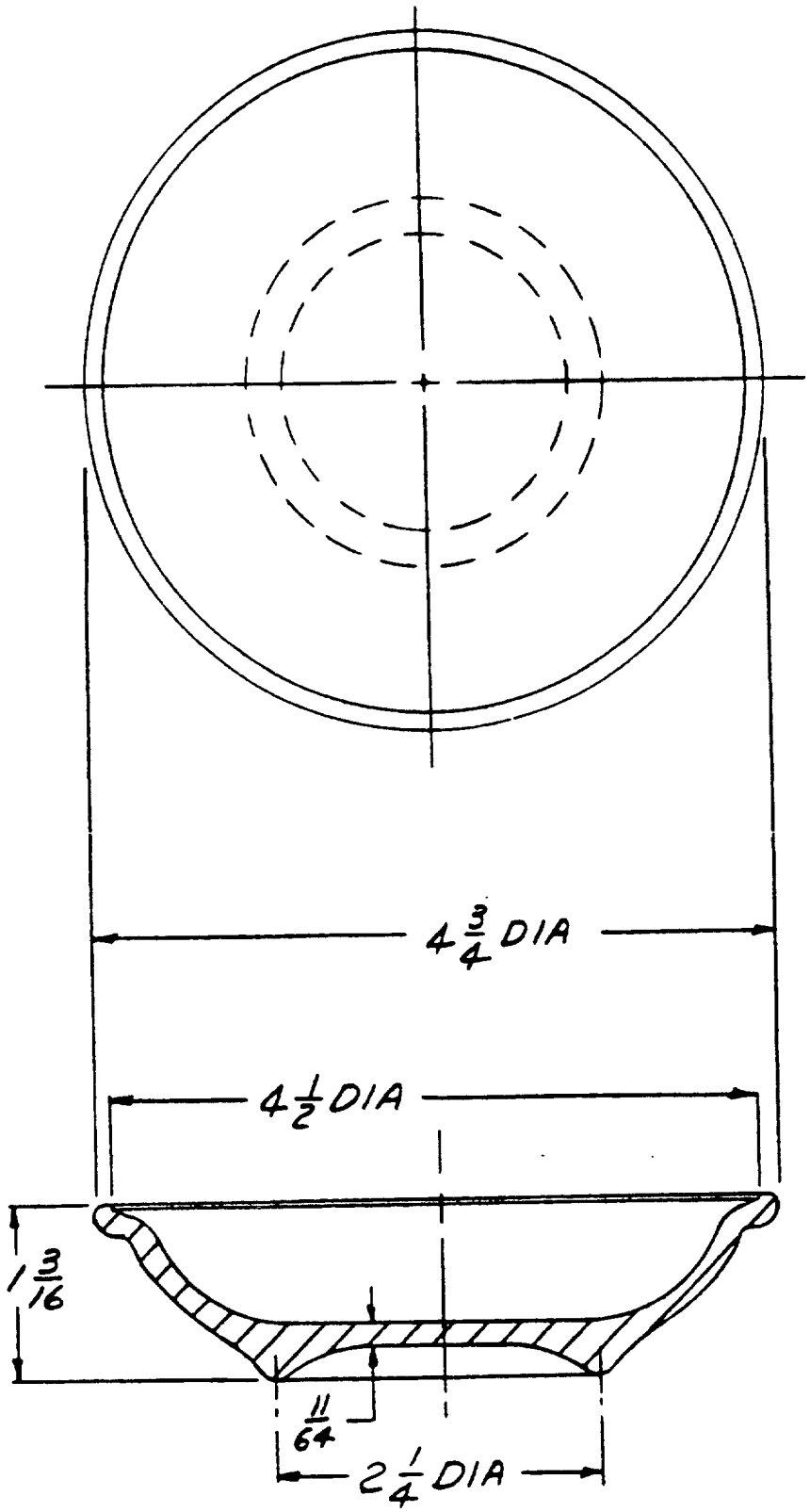
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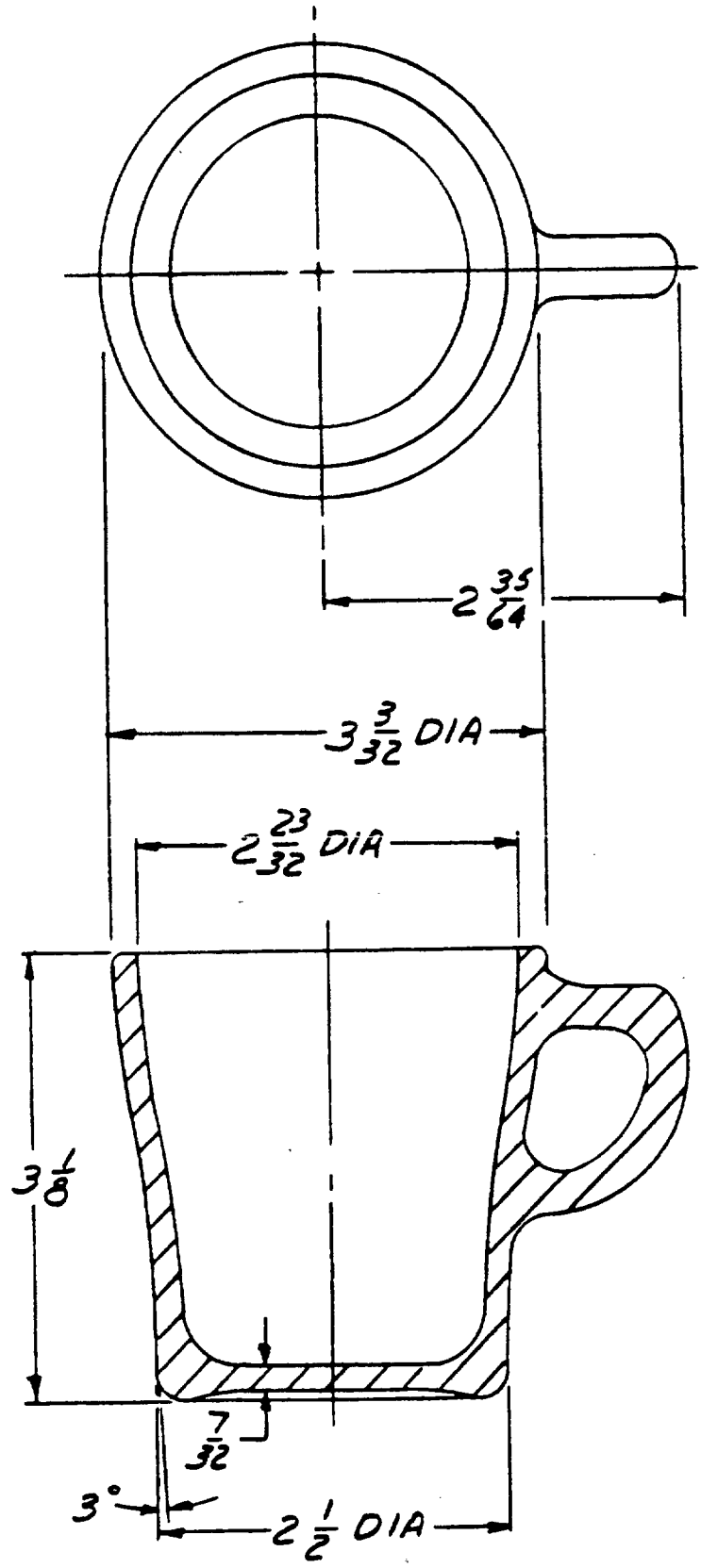
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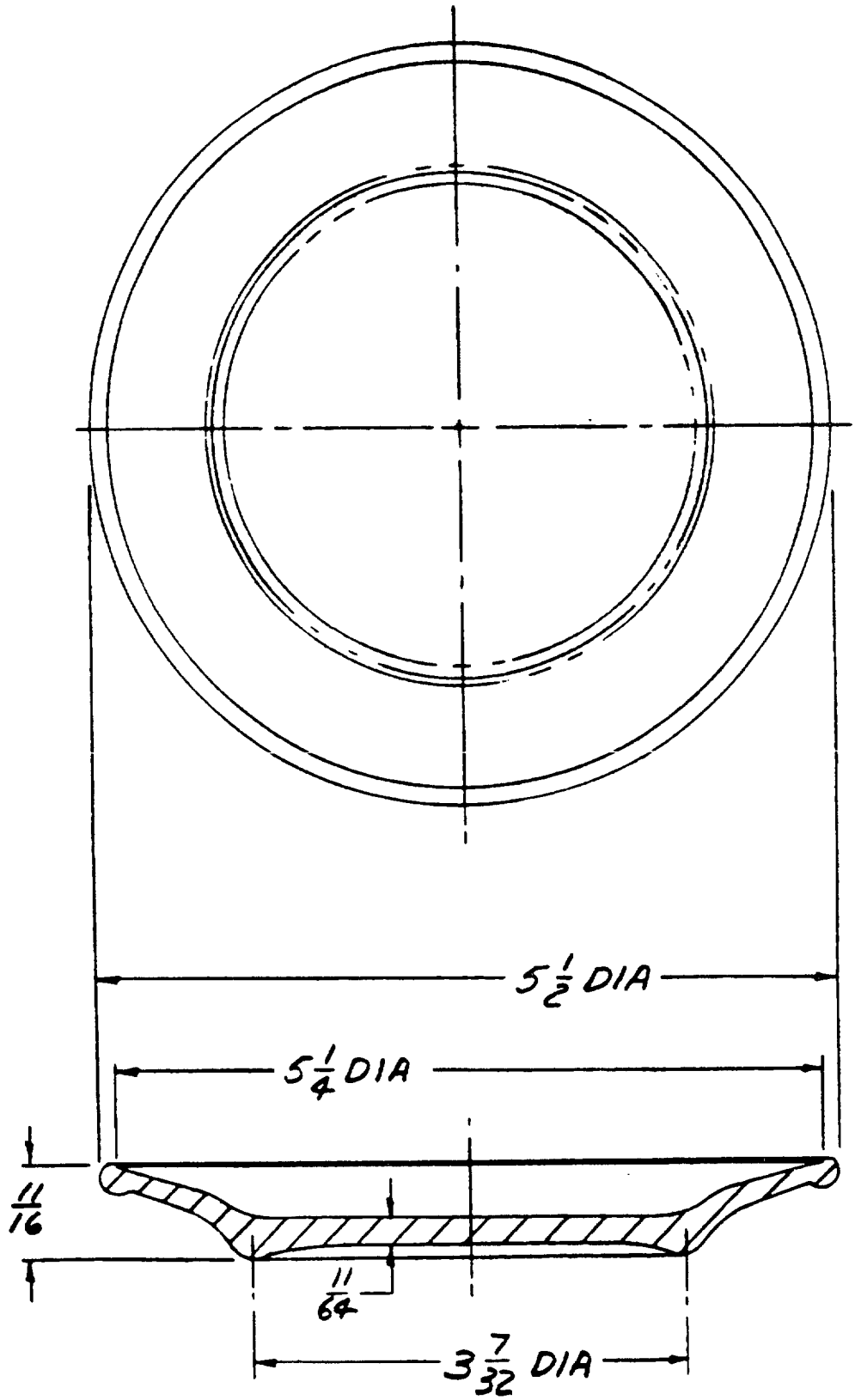
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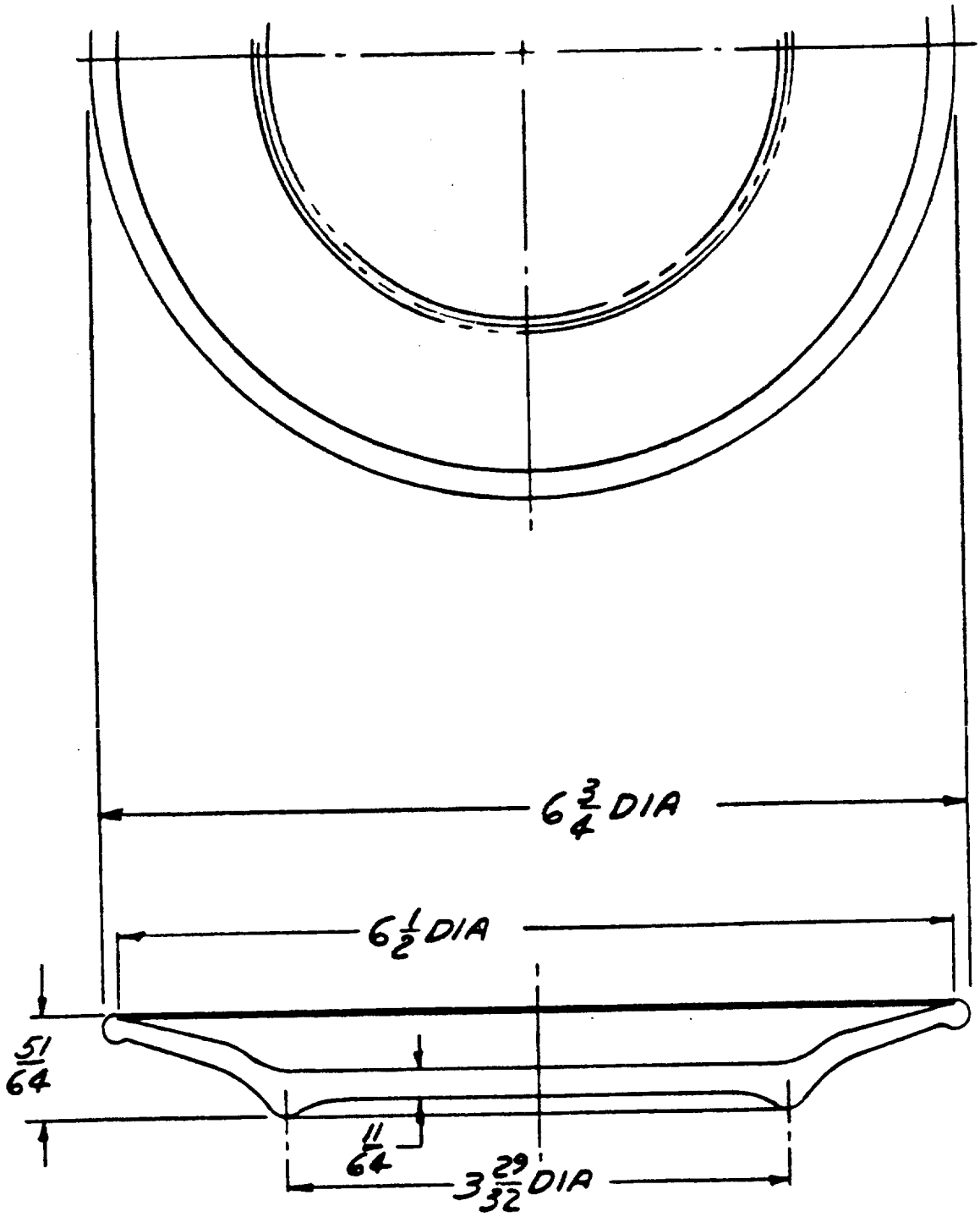
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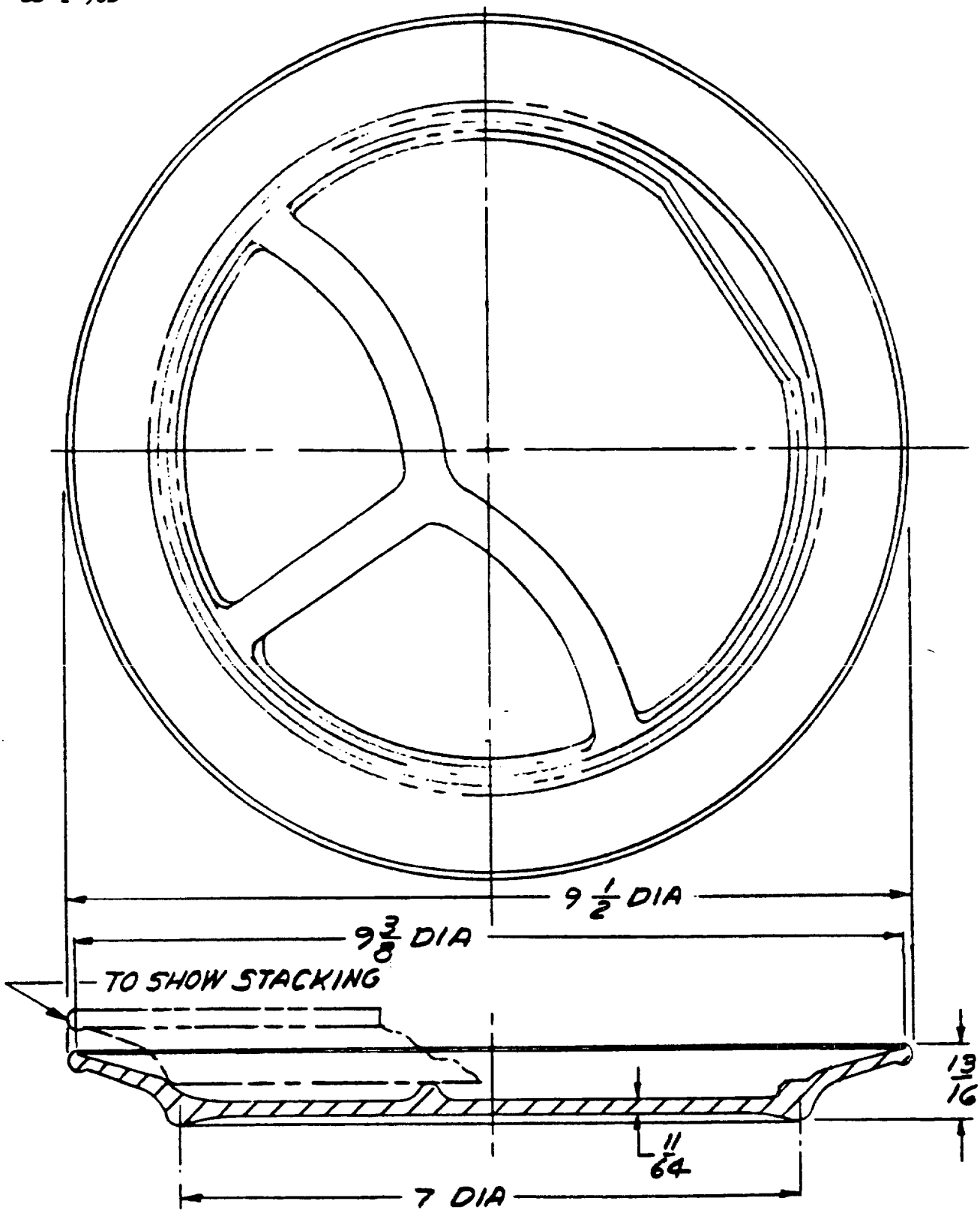
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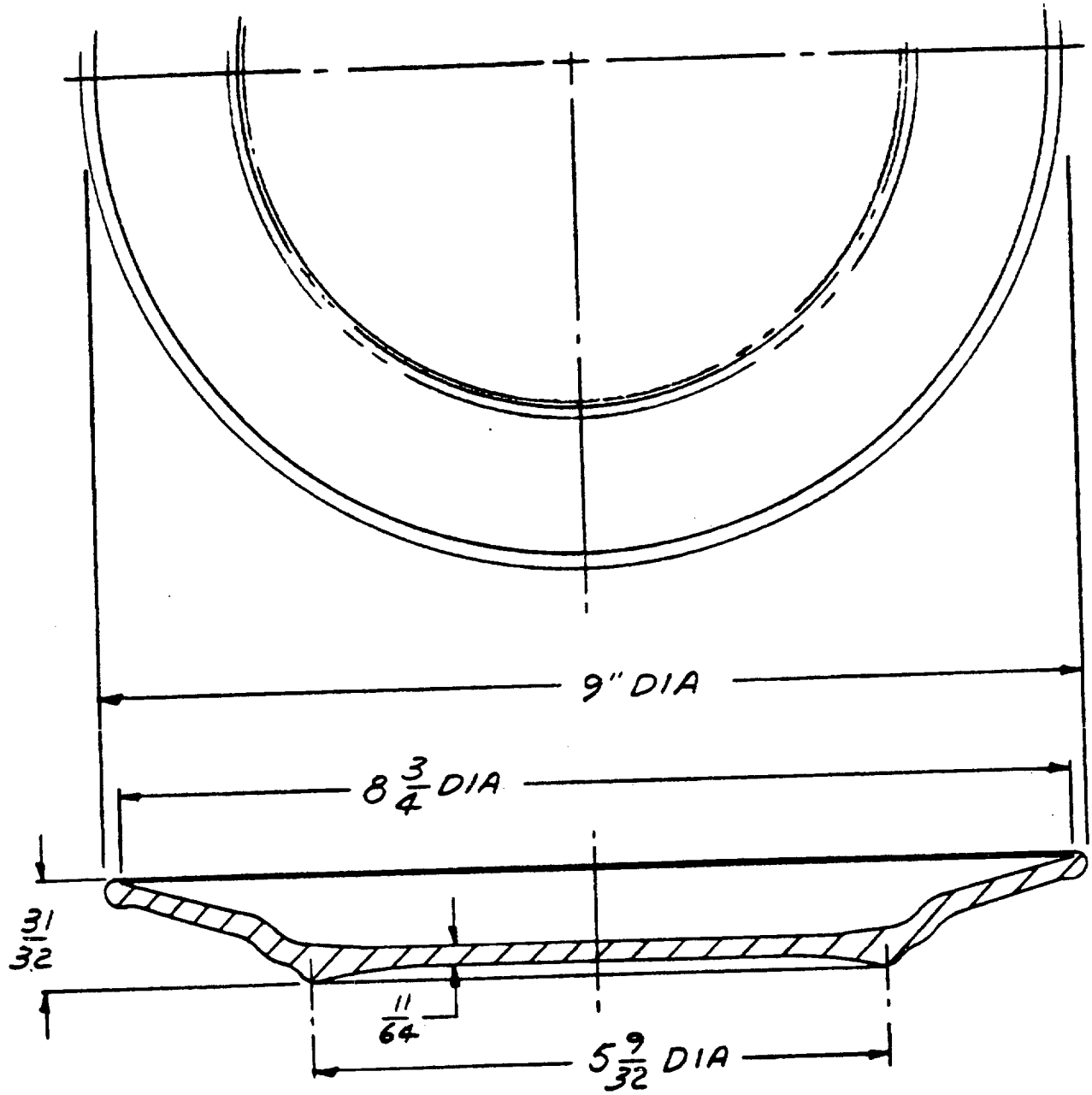
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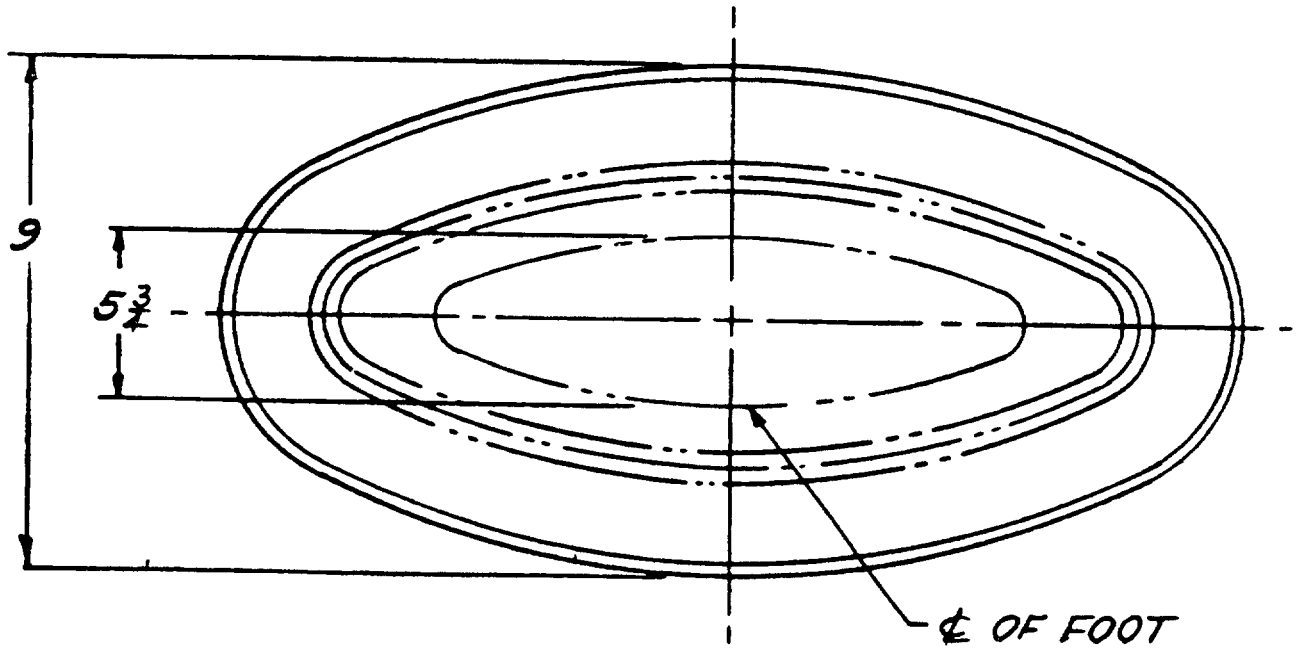
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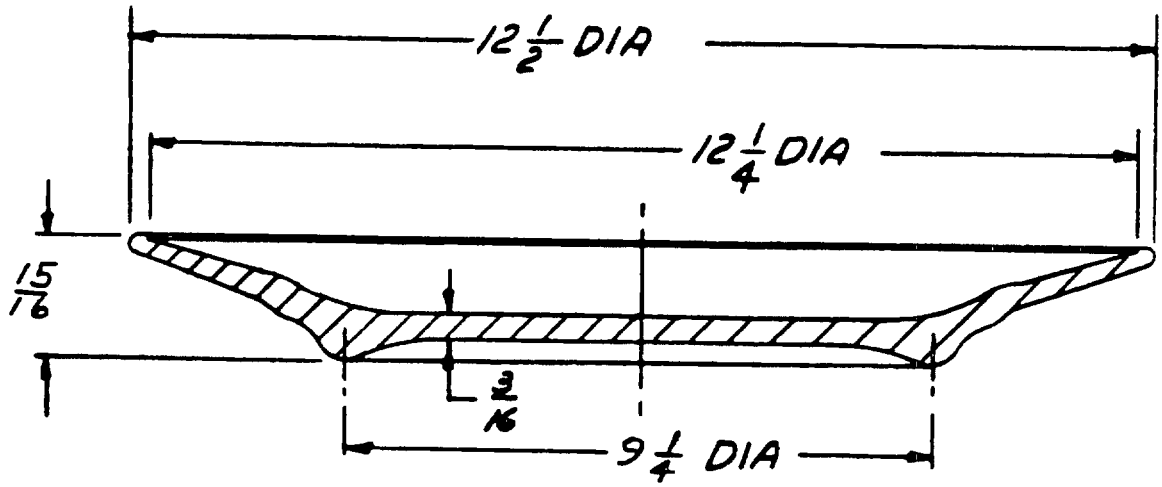
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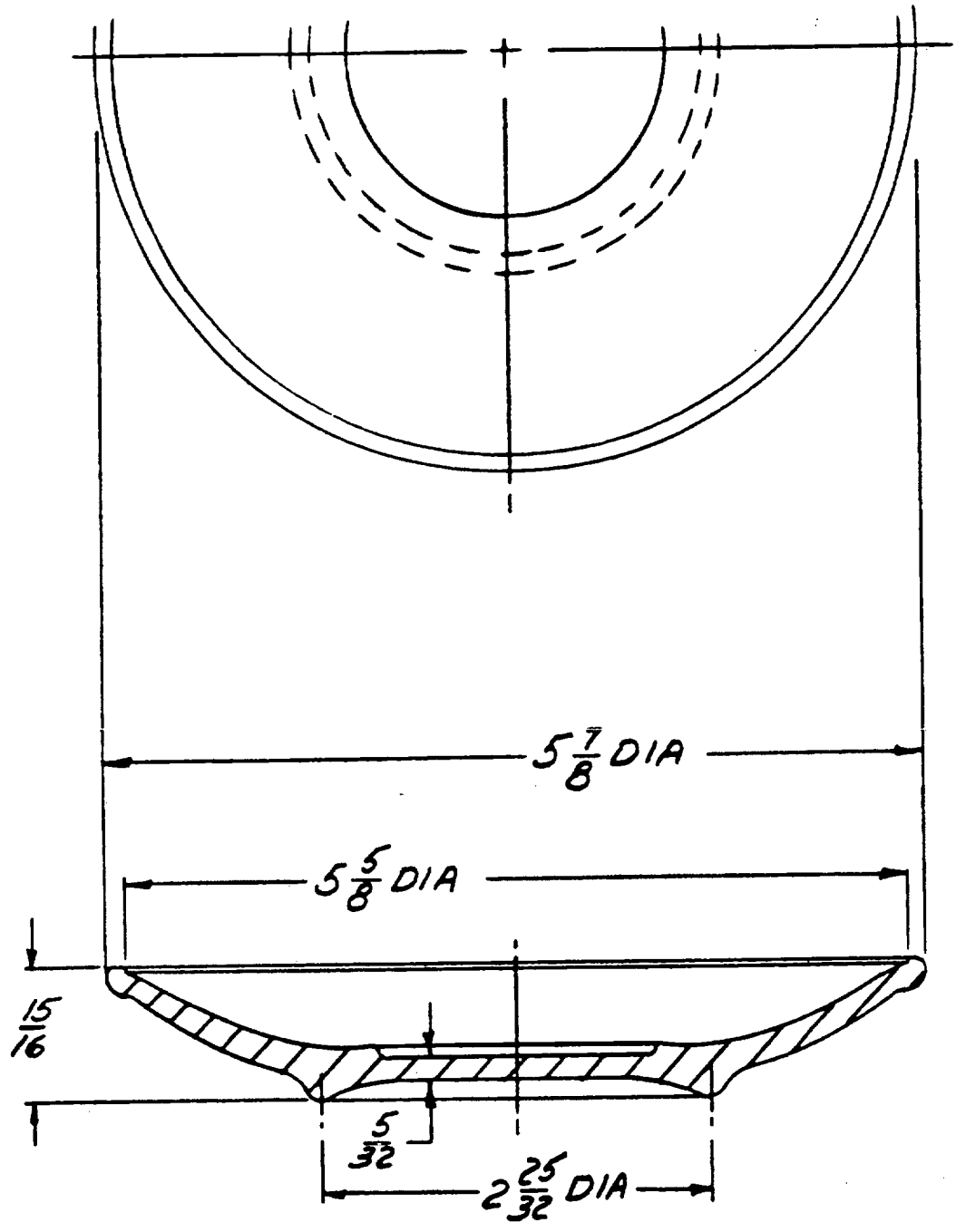
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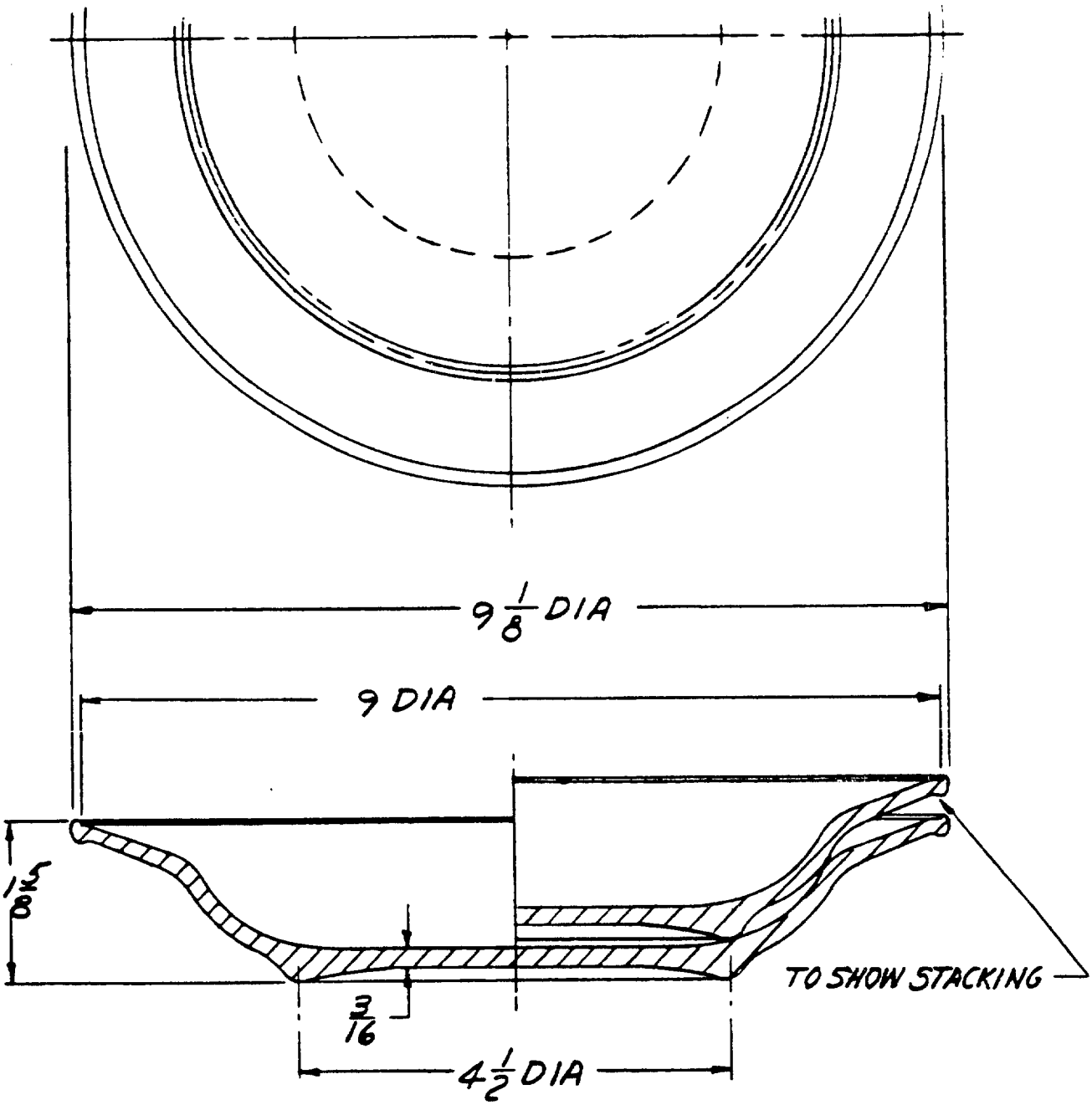
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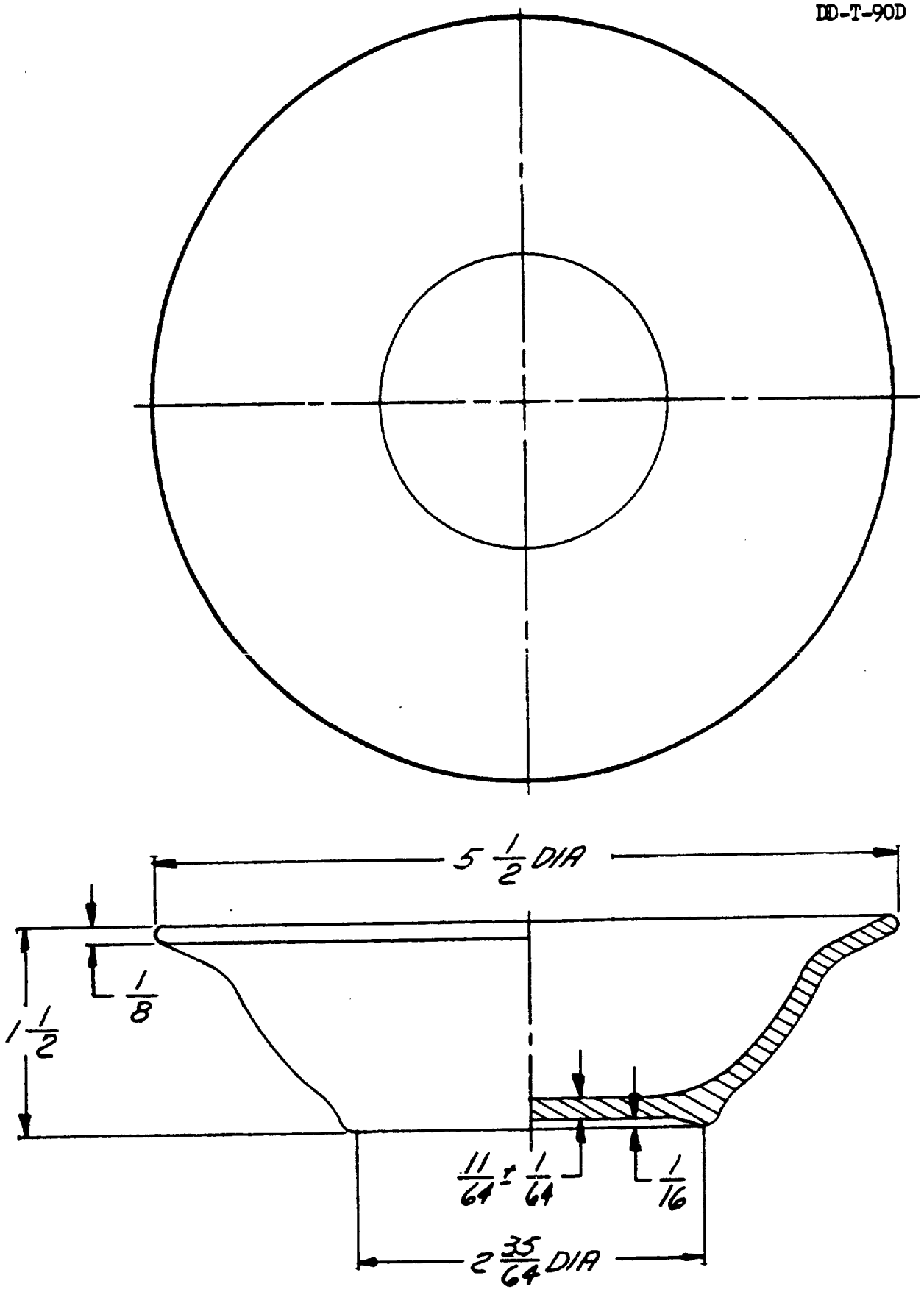
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Air Force - 84

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Army - MD
Navy - MC, MS

User activities:

Navy - CG, YD

Preparing activity:

Army - GL

Project No. 7350-0137

CIVIL AGENCIES INTEREST:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

TABLEWARE, GLASS (BOWLS, CUPS, DISHES, PLATES, PLATTERS, AND SAUCERS) DD-T-90D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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